

ITC Tut-1

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B3.

$$(1) H(X/Y) = - \sum_{x,y} P(x,y) \log_2 \left(\frac{P(x,y)}{P(y)} \right)$$

$$P(y_0) = (1-p)q + p(1-q)$$

$$P(y_1) = pq + (1-p)(1-q)$$

$$P(x_0, y_0) = (1-p)q$$

$$P(x_1, y_0) = p(1-q)$$

$$P(x_0, y_1) = (1-p)(1-q)$$

$$P(x_1, y_1) = pq$$

$$H(X,Y) = -[(1-p)q \log_2((1-p)q) + p(1-q) \log_2(p(1-q))]$$

$$\therefore p = 0.1 \Rightarrow (p = 1-q \Rightarrow q = 1-p)$$

$$H(X/Y) = -[(1-0.1)q \log_2((1-0.1)q) + 0.1(1-q) \log_2(0.1(1-q))]$$

$$H(q, 0.1) = -[(0.9q) \log_2(0.9q) + 0.1(1-q) \log_2(0.1(1-q))]$$

$$H(q, 0.2) = -[(0.8q) \log_2(0.8q) + 0.2(1-q) \log_2(0.2(1-q))]$$

$$H(q, 0.5) = -[(0.5q) \log_2(0.5q) + 0.5(1-q) \log_2(0.5(1-q))]$$

$$H(q, 0.8) = -[(0.2q) \log_2(0.2q) + 0.2(1-q) \log_2(0.2(1-q))]$$

$$H(q, 0.9) = -[(0.1q) \log_2(0.1q) + 0.1(1-q) \log_2(0.1(1-q))]$$

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The above values of q are optimal values.

For,

$$p=0.1, q=0.9 \Rightarrow H(x/y) = 0.137$$

$$p=0.2, q=0.8 \Rightarrow H(x/y) = 0.258$$

$$p=0.5, q=0.5 \Rightarrow H(x/y) = 0.50$$

$$p=0.8, q=0.2 \Rightarrow H(x/y) = 0.258$$

$$p=0.9, q=0.1 \Rightarrow H(x/y) = 0.137$$

(Ans.)

(2) (i) $H(y) = 0$ bits has same uncertainty as $y(x)$ is deterministic.

(ii) $y(x)$ is deterministic.

x gives info about $y \Rightarrow H(y/x) = 0$ bits.

(iii) $H(x, y) = H(x) + H(y) = 8$ bits (Joint probability).

(iv) If $y(x)$ is not invertible, $\text{entropy}(y) < \text{entropy}(x)$
Implies there is some uncertainty. $H(y) > 0$ bits.

(v) With not invertible $y(x)$,

$$H(x/y) > 0 \text{ bits}$$

X - Axis :

Y - Axis :

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Date :

$H(X)$

X axis 1cm = 0.1 units

Y axis 1cm = 0.1 units

0.6
0.5
0.4
0.3
0.2
0.1

0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0

$q \rightarrow$

